2 BP

D-subsets

$$\begin{cases} X_{2j} = X_{1j} - X_{13} & j = 4, \dots, n-1 \\ X_{2n} = -X_{13} \end{cases}$$

assume no multiples

$$\sum_{3 \le j < l \le n-1} \frac{e_{1j\,1l}}{X_{1j} X_{1l}} + \sum_{\substack{3 \le j \le n-1 \\ 4 \le l \le n}} \frac{e_{1j\,2l}}{X_{1j} X_{2l}} + \sum_{4 \le j < l \le n} \frac{e_{2j\,2l}}{X_{2j} X_{2l}}$$

$$= \frac{1}{X_{13}} \sum_{j=4}^{n-1} \frac{e_{13\,1j}}{X_{1j}} + \sum_{4 \le j < l \le n-1} \frac{e_{1j\,1l}}{X_{1j} X_{1l}} + \frac{1}{X_{13}} \sum_{j=4}^{n-1} \frac{e_{13\,2j}}{X_{2j}} + \frac{1}{X_{2n}} \sum_{j=4}^{n-1} \frac{e_{2n\,1j}}{X_{1j}} + \sum_{j,l=4}^{n-1} \frac{e_{1j\,2l}}{X_{1j} X_{2l}}$$

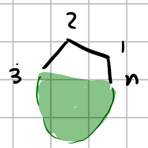
$$\frac{e_{13\,2n}}{X_{13} X_{2n}} + \frac{1}{X_{2n}} \sum_{j=4}^{n-1} \frac{e_{2n\,2j}}{X_{2j}} + \sum_{4 \le j < l \le n-1} \frac{e_{2j\,2l}}{X_{2j} X_{2l}}$$

---

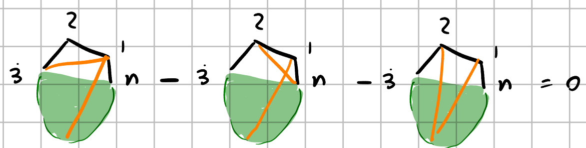
multiply by X_{1j}, send $X_{1j} \to 0$ $\quad (j \in \{4, \dots, n-1\})$ $\quad X_{1j} = 0 \Rightarrow X_{2j} = -X_{13}$

$$\rightarrow \frac{e_{13\,1j}}{X_{13}} + \sum_{l \ne j} \frac{e_{1j\,1l}}{X_{1l}} + \frac{e_{2n\,1j}}{X_{2n}} + \sum_{l} \frac{e_{1j\,2l}}{X_{2l}} = 0 \qquad \rightarrow \sum_{l \ne j} \frac{e_{1j\,2l}}{X_{2l}} + \frac{e_{1j\,2j}}{X_{2j}}$$

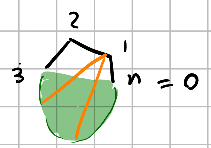
$$\rightarrow \frac{1}{X_{13}} \left(e_{13\,1j} - e_{2n\,1j} \right) + \sum_{l \ne j} \frac{e_{1j\,1l}}{X_{1l}} + \sum_{l \ne 0} \frac{e_{1j\,2l}}{X_{1l} - X_{13}} - \frac{e_{1j\,2j}}{X_{13}}$$

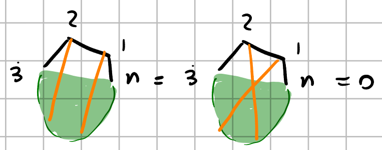


- Residue on X_{13}: $\quad e_{13\,1j} - e_{2n\,1j} - e_{1j\,2j}$



- Residue on $X_{1l} = 0 \ (l \ne j)$: $\boxed{e_{1j\,1l} = 0}$
- Residue on $X_{1l} = X_{13}$ $\ (X_{2l} = 0)$: $\boxed{e_{1j\,2l} = 0}$





Same idea with X_{2j} as residue!

$$\frac{e_{13\,2j}}{X_{13}} + \frac{e_{2n\,2j}}{X_{2n}} + \sum_{l \ne j} \frac{e_{2j\,1l}}{X_{1l}} + \frac{e_{2j\,1j}}{X_{1j}} + \sum_{l \ne j} \frac{e_{2j\,2l}}{X_{2l}}$$

$X_{13} = -X_{2n}$

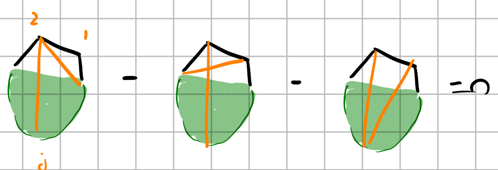
$X_{1l} = X_{2l} - X_{2n}$

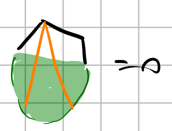
$$\rightarrow \frac{e_{2n\,2j} - e_{13\,2j} - e_{2j\,1j}}{X_{2n}} + \sum_{l \ne j} \frac{e_{2j\,1l}}{X_{2l} - X_{2n}} + \sum_{l \ne j} \frac{e_{2j\,2l}}{X_{2l}} = 0$$

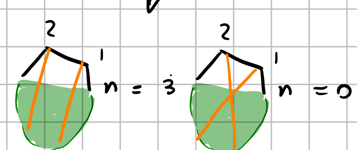
$X_{2n} = 0$ residue $\qquad X_{2l} = X_{2n}$ residue $\qquad X_{2l} = 0$ residue

$\boxed{e_{2n\,2j} - e_{13\,2j} - e_{2j\,1j} = 0}$ $\qquad e_{2j\,1l} = 0 \quad j \ne l$ $\qquad e_{2j\,2l} = 0$

same as before



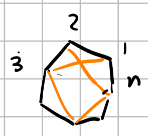


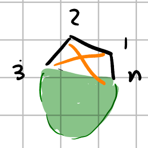


start by imposing $X_{2n} = -X_{13}$

$$\rightarrow \quad D_{\text{subset}} = -\frac{a_{13\,2n}}{X_{13}^2} + O(X_{13}^{-1})$$

Residue at $X_{13}=\infty$ $\rightarrow$ $\boxed{a_{13\,2n} = 0}$ $\nearrow$ new


$= 0$


$= 0$

1-zeros kill everything at $n=7$

3BP

$$DS = \sum_{3\le j_1<j_2<j_3\le n-1} \frac{a_{1j_1\,1j_2\,1j_3}}{X_{1j_1}X_{1j_2}X_{1j_3}} + \sum_{4\le j_1<j_2<j_3\le n}\frac{a_{2j_1\,2j_2\,2j_3}}{X_{2j_1}X_{2j_2}X_{2j_3}} + \sum_{3\le j_1<j_2\le n-1}\sum_{j_3=4}^{n}\frac{a_{1j_1\,1j_2\,2j_3}}{X_{1j_1}X_{1j_2}X_{2j_3}} + \sum_{4\le j_1<j_2\le n}\sum_{j_3=3}^{n-1}\frac{a_{2j_1\,2j_2\,1j_3}}{X_{2j_1}X_{2j_2}X_{1j_3}}$$

$$= \frac{1}{X_{13}}\underbrace{\sum_{j_1<j_2}}\frac{a_{13\,1j_1\,1j_2}}{X_{1j_1}X_{1j_2}} + \sum_{j_1<j_2<j_3}\frac{a_{1j_1\,1j_2\,1j_3}}{X_{1j_1}X_{1j_2}X_{1j_3}} + \frac{1}{X_{2n}}\sum_{j_1<j_2}\frac{a_{2j_1\,2j_2\,2n}}{X_{2j_1}X_{2j_2}} + \sum_{j_1<j_2<j_3}\frac{a_{2j_1\,2j_2\,2j_3}}{X_{2j_1}X_{2j_2}X_{2j_3}}$$

$\hookrightarrow$ assume bounds are always $4, \dots, n-1$

$$+ \frac{1}{X_{13}}\sum_{j_1 j_2}\frac{a_{13\,1j_1\,2j_2}}{X_{1j_1}X_{2j_2}} + \frac{1}{X_{13}X_{2n}}\sum_{j}\frac{a_{13\,2n\,1j}}{X_{1j}} + \frac{1}{X_{2n}}\sum_{j_1<j_2}\frac{a_{1j_1\,1j_2\,2n}}{X_{1j_1}X_{1j_2}} + \sum_{j_1<j_2}\sum_{j_3}\frac{a_{1j_1\,1j_2\,2j_3}}{X_{1j_1}X_{1j_2}X_{2j_3}}$$

$$+ \frac{1}{X_{2n}}\sum_{j_1 j_2}\frac{a_{2n\,1j_1\,2j_2}}{X_{1j_1}X_{2j_2}} + \frac{1}{X_{13}X_{2n}}\sum_{j}\frac{a_{13\,2n\,2j}}{X_{2j}} + \frac{1}{X_{13}}\sum_{j_1<j_2}\frac{a_{2j_1\,2j_2}}{X_{2j_1}X_{2j_2}} + \sum_{j_1<j_2}\sum_{j_3}\frac{a_{2j_1\,2j_2\,1j_3}}{X_{2j_1}X_{2j_2}X_{1j_3}}$$

start by setting $X_{2n} = -X_{13}$

$$DS = -\frac{1}{X_{13}^2}\left(\sum_j \frac{a_{13\,2n\,2j}}{X_{2j}} + \sum_j \frac{a_{13\,2n\,1j}}{X_{1j}}\right) + \frac{1}{X_{13}}\left(\sum_{j_1 j_2}\frac{a_{13\,1j_1\,2j_2}}{X_{1j_1}X_{2j_2}} - \sum_{j_1<j_2}\frac{a_{1j_1\,1j_2\,2n}}{X_{1j_1}X_{1j_2}} + \right.$$

$$\left. - \sum_{j_1 j_2}\frac{a_{2n\,1j_1\,2j_2}}{X_{1j_1}X_{2j_2}} + \sum_{j_1<j_2}\frac{a_{2j_1\,2j_2}}{X_{2j_1}X_{2j_2}} + \sum_{j_1<j_2}\frac{a_{13\,1j_1\,1j_2}}{X_{1j_1}X_{1j_2}} - \sum_{j_1<j_2}\frac{a_{2j_1\,2j_2\,2n}}{X_{2j_1}X_{2j_2}}\right)$$

$$+ \sum_{j_1<j_2<j_3}\frac{a_{1j_1\,1j_2\,1j_3}}{X_{1j_1}X_{1j_2}X_{1j_3}} + \sum_{j_1<j_2<j_3}\frac{a_{2j_1\,2j_2\,2j_3}}{X_{2j_1}X_{2j_2}X_{2j_3}} + \sum_{j_1<j_2}\sum_{j_3}\frac{a_{1j_1\,1j_2\,2j_3}}{X_{1j_1}X_{1j_2}X_{2j_3}} + \sum_{j_1<j_2}\sum_{j_3}\frac{a_{2j_1\,2j_2\,1j_3}}{X_{2j_1}X_{2j_2}X_{1j_3}}$$

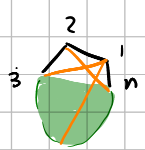
---

① multiply by X_{13}^2, $X_{13}\to 0$

$X_{2j} = X_{1j}$

$$\sum_j \frac{a_{13\,2n\,2j}}{X_{2j}} + \sum_j \frac{a_{13\,2n\,1j}}{X_{1j}} = 0 \quad \rightarrow \quad \sum_j \frac{a_{13\,2n\,2j}}{X_{1j}} + \frac{a_{13\,2n\,1j}}{X_{1j}} = 0$$

$$\Rightarrow \quad \boxed{a_{13\,2n\,2j} + a_{13\,2n\,1j} = 0}$$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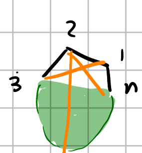

$= 0$

Residue at X_{1j} ← do this first

$$-\frac{1}{X_{13}^2}a_{13\,2n\,1j} + \frac{1}{X_{13}}\left(\sum_{\ell}\frac{a_{13\,1j\,2\ell}}{X_{2\ell}} - \sum_{\ell\neq j}\frac{a_{1j\,1\ell\,2n}}{X_{1\ell}} + \right.$$

$$\left. - \sum_{\ell}\frac{a_{2n\,1j\,2\ell}}{X_{2\ell}} + \sum_{\ell\neq j}\frac{a_{13\,1j\,1\ell}}{X_{1\ell}}\right) + \sum_{\substack{j_1<j_2\\ j_1,j_2\neq j}}\frac{a_{1j_1\,1j_2\,1j}}{X_{1j_1}X_{1j_2}} + \sum_{\ell\neq j}\sum_{k}\frac{a_{1j\,1\ell\,2k}}{X_{1\ell}X_{2k}} + \sum_{j_1<j_2}\frac{a_{2j_1\,2j_2\,1j}}{X_{2j_1}X_{2j_2}}$$

$$= -\frac{1}{X_{13}^2}a_{13\,2n\,1j} + \frac{1}{X_{13}}\left[\sum_{\ell\neq j}\left(\frac{1}{X_{1\ell}}\left(a_{13\,1j\,1\ell} - a_{1j\,1\ell\,2n}\right) + \frac{1}{X_{2\ell}}\left(a_{13\,1j\,2\ell} - a_{2n\,1j\,2\ell}\right)\right)\right.$$

$$\left. + \frac{1}{X_{2j}}\left(a_{13\,1j\,2j} - a_{2n\,1j\,2j}\right)\right] + \sum_{\substack{j_1<j_2\\ j_1,j_2\neq j}}\left(\frac{a_{1j_1\,1j_2\,1j}}{X_{1j_1}X_{1j_2}} + \frac{a_{2j_1\,2j_2\,1j}}{X_{2j_1}X_{2j_2}}\right) + \sum_{\ell\neq j}\sum_{k\neq\ell}\frac{a_{1j\,1\ell\,2k}}{X_{1\ell}X_{2k}} + \frac{1}{X_{2j}}\sum_{\ell\neq j}\frac{a_{1j\,1\ell\,2k}}{X_{1\ell}}$$

- easier thing: multiply by $X_{1j_1}X_{1j_2}X_{1j_3}$, send all to zero:

$\rightarrow a_{1j_1\,1j_2\,1j_3} = 0$

- next multiply by $X_{1j_1}X_{1j_2}$, send all to zero:

$$\sum_{\ell\neq j_1,j_2}\frac{a_{1j_1\,1j_2\,2\ell}}{X_{2\ell}} + \frac{a_{1j_1\,1j_2\,2j_1}}{\underbrace{X_{2j_1}}_{=-X_{13}}} + \frac{a_{1j_1\,1j_2\,2j_2}}{\underbrace{X_{2j_2}}_{=-X_{13}}} + \frac{a_{13\,1j_1\,1j_2} - a_{2n\,1j_1\,1j_2}}{X_{13}} = 0$$

$$= \sum_{\ell\neq j_1,j_2}\frac{a_{1j_1\,1j_2\,2\ell}}{X_{2\ell}} + \frac{1}{X_{13}}\left(a_{13\,1j_1\,1j_2} - a_{2n\,1j_1\,1j_2} - a_{1j_1\,1j_2\,2j_1} - a_{1j_1\,1j_2\,2j_1}\right) = 0$$

so $\boxed{a_{1j_1\,1j_2\,2\ell} = 0}$ with $j_1\neq j_2\neq\ell$ ⟶ $a_{1j_1\,1j_2\,2j_3} = 0$ if $j_1\neq j_2\neq j_3$

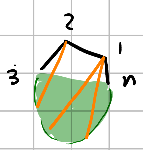
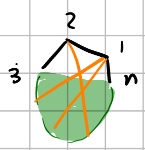
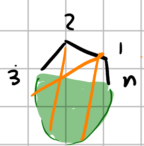
$\boxed{a_{13\,1j_1\,1j_2} - a_{2n\,1j_1\,1j_2} - a_{1j_1\,1j_2\,2j_1} - a_{1j_1\,1j_2\,2j_1} = 0 \quad j_1\neq j_2}$

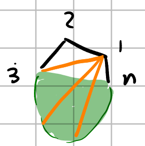
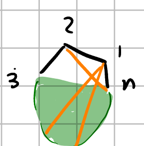
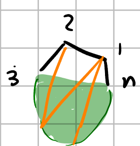
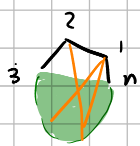
↓

$a_{i_1j_1\,i_2j_2\,i_3j_3} = 0$

$i_1,i_2,i_3\in\{1,2\}$

$j_1\neq j_2\neq j_3$

 =  =  = 0

 −  −  −  = 0

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look at residue of X_{1j} :

↓

$$-\frac{1}{X_{13}^2}\, a_{13\,2n\,1j} + \frac{1}{X_{13}}\left(\sum_{\ell} \frac{a_{13\,1j\,2\ell}}{X_{2\ell}} - \sum_{\ell\neq j} \frac{a_{1j\,1\ell\,2n}}{X_{1\ell}} + \right.$$

$$\left. - \sum_{\ell} \frac{a_{2n\,1j\,2\ell}}{X_{2\ell}} + \sum_{\ell\neq j} \frac{a_{13\,1j\,1\ell}}{X_{1\ell}} \right) + \sum_{\ell\neq j}\sum_{k} \frac{a_{1j\,1\ell\,2k}}{X_{1\ell}X_{2k}} + \sum_{j_1<j_2} \frac{a_{2j_1\,2j_2\,1j}}{X_{2j_1}X_{2j_2}}$$

$$= -\frac{1}{X_{13}^2}\, a_{13\,2n\,1j} + \frac{1}{X_{13}}\left(\frac{a_{13\,1j\,2\ell}}{X_{2j}} - \frac{a_{2n\,1j\,2\ell}}{X_{2j}} + \sum_{\ell\neq j} \frac{a_{13\,1j\,1\ell} - a_{2n\,1j\,1\ell}}{X_{1\ell}} + \sum_{\ell\neq j} \frac{a_{13\,1j\,2\ell} - a_{2n\,1j\,2\ell}}{X_{2\ell}} \right)$$

$$\sum_{\ell\neq k\neq j} \frac{a_{1j\,1\ell\,2k}}{X_{1\ell}X_{2k}} + \sum_{\ell\neq j} \frac{a_{1j\,1\ell\,2\ell}}{X_{1\ell}X_{2\ell}} + \sum_{\ell\neq j} \frac{a_{1j\,1\ell\,2j}}{X_{1\ell}X_{2j}} + \sum_{\substack{j_1<j_2\\ j_1,j_2\neq j}} \frac{a_{2j_1\,2j_2\,1j}}{X_{2j_1}X_{2j_2}} + \sum_{\ell\neq j} \frac{a_{1j\,2j\,2\ell}}{X_{2j}X_{2\ell}}$$

vanishes

$(1j_1, 1j_2)$ ↓

$\boxed{a_{1j_1\,1j_2\,2\ell} = 0}$ with $j_1 \neq j_2 \neq \ell$

$\boxed{a_{13\,1j_1\,1j_2} - a_{2n\,1j_1\,1j_2} - a_{1j_1\,1j_2\,2j_1} - a_{1j_1\,1j_2\,2j_1} = 0 \quad j_1 \neq j_2}$

$(2j_1, 2j_2)$ ↓

$a_{2j_1\,2j_2\,1\ell} = 0 \quad j_1 \neq j_2 \neq \ell$

$\boxed{a_{2n\,2j_1\,2j_2} - a_{13\,2j_1\,2j_2} - a_{2j_1\,2j_2\,1j_1} - a_{2j_1\,2j_2\,1j_2} = 0} \quad (j_1 \neq j_2)$

impose $X_{2j} = -X_{13}$

$$\frac{1}{X_{13}^2}\left(-a_{13\,2n\,1j} - a_{13\,1j\,2\ell} + a_{2n\,1j\,2\ell} \right) +$$

$$\frac{1}{X_{13}} \sum_{\ell\neq j}\left(\frac{a_{13\,1j\,1\ell} - a_{2n\,1j\,1\ell}}{X_{1\ell}} + \frac{a_{13\,1j\,2\ell} - a_{2n\,1j\,2\ell}}{X_{2\ell}} - \frac{a_{1j\,1\ell\,2j}}{X_{1\ell}} - \frac{a_{1j\,2j\,2\ell}}{X_{2\ell}} \right)$$

$$+ \sum_{\ell\neq k\neq j} \frac{a_{1j\,1\ell\,2k}}{X_{1\ell}X_{2k}} + \sum_{\ell\neq j} \frac{a_{1j\,1\ell\,2\ell}}{X_{1\ell}X_{2\ell}}$$

$\rightarrow \boxed{-a_{13\,2n\,1j} - a_{13\,1j\,2\ell} + a_{2n\,1j\,2\ell} = 0}$

$\rightarrow$ residue at X_{13} : $\displaystyle\sum_{\ell\neq j}\left(\frac{a_{13\,1j\,1\ell} - a_{2n\,1j\,1\ell}}{X_{1\ell}} + \frac{a_{13\,1j\,2\ell} - a_{2n\,1j\,2\ell}}{X_{1\ell}} - \frac{a_{1j\,1\ell\,2j}}{X_{1\ell}} - \frac{a_{1j\,2j\,2\ell}}{X_{1\ell}} \right)$

$\rightarrow \boxed{a_{13\,1j\,1\ell} - a_{2n\,1j\,1\ell} + a_{13\,1j\,2\ell} - a_{2n\,1j\,2\ell} - a_{1j\,1\ell\,2j} - a_{1j\,2j\,2\ell}} = 0$

$\rightarrow$ residue at $X_{1\ell} = 0$: $\displaystyle \frac{a_{13\,1j\,1\ell} - a_{2n\,1j\,1\ell}}{X_{13}} + \sum_{\ell\neq k\neq j} \frac{a_{1j\,1\ell\,2k}}{X_{2k}} + \frac{a_{1j\,1\ell\,2\ell}}{X_{2\ell}}$ $\rightarrow -X_{13}$

then residue at X_{13}: $\boxed{a_{13\,1j\,1\ell} - a_{2n\,1j\,1\ell} - a_{1j\,1\ell\,2\ell} = 0}$

$\boxed{a_{1j\,1\ell\,2k} = 0} \quad j \neq k \neq \ell$

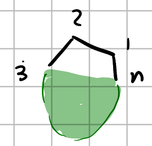
eqs with both 13 and 2n:

$$\boxed{a_{13\,2n\,2j} + a_{13\,2n\,1j} = 0}$$

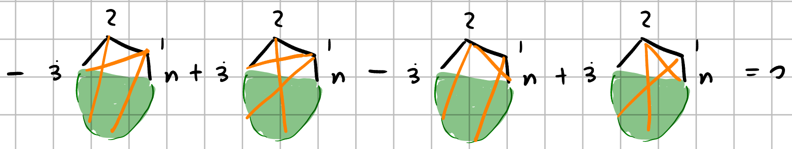
$$\boxed{-a_{13\,2n\,1j} - a_{13\,1j\,2l} + a_{2n\,1j\,2l} = 0}$$

using $2j_1$ instead of $1j_1$

$$\boxed{-a_{2n\,13\,2j} - a_{2n\,2j\,1l} + a_{13\,2j\,1l} = 0}$$

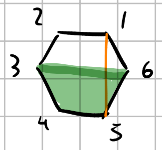


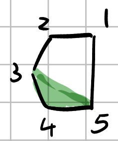
$$\Rightarrow \quad -a_{13\,1j\,2l} + a_{2n\,1j\,2l} - a_{2n\,2j\,1l} + a_{13\,2j\,1l} = 0$$



subconstraints

take away anything that affects to 6





$24 = 14 - 13$
$25 = 15 - 13$
$26 = -13$

$\xrightarrow{X_{15}=0}$

$24 = 14 - 13$
$25 = -13$

2-zero for $n = 6$

$35 = 15 - 14$
$36 = -14$
$25 = 35 + 24$
$26 = 36 + 24$

$\xrightarrow{15=0}$

$35 = -14$
$25 = 35 + 24$
$= 24 - 14$

$\downarrow +2$

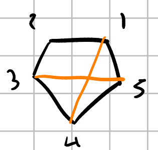
$\boxed{24 = 14 - 13 \quad 25 = -13}$

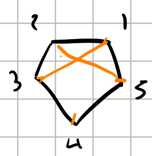
2-zero $n = 7$

$35 = 15 - 14$
$36 = 16 - 14$
$37 = 17 - 14$
$25 = 35 + 24$
$26 = 36 + 24$
$27 = 37 + 24$

$\xrightarrow{16=0}$

$35 = 15 - 14$
$36 = -14$
$25 = 35 + 24$
$26 = 36 + 24$





---

try rotated zeros

+0

$24 = 14 - 13$
$25 = 15 - 13$
$- 26 = -13$

$\xrightarrow{X_{15}=0}$

$24 = 14 - 13$
$25 = -13$

$\downarrow +1$

+1

$35 = 25 - 24$
$- 36 = 26 - 24$
$31 = -24$

$\rightarrow$

$35 = 25 - 24$
$31 = -24$

1-zero for $n=5$

$\downarrow +1$

+2

$- 46 = 36 - 35$
$41 = 31 - 35$
$42 = -35$

$\rightarrow$

$41 = 31 - 35$
$42 = -35$

✓

+1

+3

$51 = 41 - 46$
$52 = 42 - 46$
$53 = -46$

$\xrightarrow{X_{13}=0}$

$\begin{cases} X_{46} = X_{41} \\ X_{52} = X_{42} - X_{46} \\ X_{53} = -X_{46} \end{cases} \rightarrow \begin{cases} - X_{46} = X_{41} \\ X_{52} = X_{42} - X_{41} \\ X_{53} = -X_{41} \end{cases} \rightarrow$

$X_{52} = X_{42} - X_{41}$
$X_{53} = -X_{41}$

+4

$62 = 52 - 51$
$63 = 53 - 51$
$64 = -51$

$\xrightarrow{15=0}$

$62 = 52$
$63 = 53$
$64 = 0$

NO

(ok because we can't do the iterative here)

+5

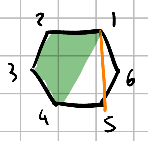
$13 = 63 - 62$
$14 = 64 - 62$
$15 = -62$

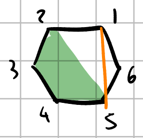
$\xrightarrow{15=0}$

$62 = 0$
$13 = 63$
$16 = 64$

NO

(ok because we can't do the iterative here)

+4 

+5 

the problem here is that 13 cannot be one of propagator in the $X_{..} = - X_{..}$ equation

---

Born-free: $(n=6)$

+0 35 36 / 46 | +1 46 41 / 51 | +2 51 52 / 62 | +3 62 63 / 14 |

26, 36, 46 are the chords crossing X_{15}

$$\underset{X_{15}}{\mathrm{Res}}\, B_6 = B_5(1,2,3,4,5) + \frac{1}{X_{26}}\left(\frac{e_{13\,15\,26}}{X_{13}} + \frac{e_{14\,15\,26}}{X_{14}} + \frac{e_{24\,15\,26}}{X_{24}} + \frac{e_{25\,15\,26}}{X_{25}} + \frac{e_{35\,15\,26}}{X_{35}}\right)$$

$$+ \frac{1}{X_{36}}\left(\frac{e_{13\,15\,36}}{X_{13}} + \frac{e_{14\,15\,36}}{X_{14}} + \frac{e_{24\,15\,36}}{X_{24}} + \frac{e_{25\,15\,36}}{X_{25}} + \frac{e_{35\,15\,36}}{X_{35}}\right)$$

$$+ \frac{1}{X_{46}}\left(\frac{e_{13\,15\,46}}{X_{13}} + \frac{e_{14\,15\,46}}{X_{14}} + \frac{e_{24\,15\,46}}{X_{24}} + \frac{e_{25\,15\,46}}{X_{25}} + \frac{e_{35\,15\,46}}{X_{35}}\right)$$

$$+ \cancel{\frac{e_{26\,36\,15}}{X_{26}X_{36}}} + \cancel{\frac{e_{26\,46\,15}}{X_{26}X_{46}}} + \cancel{\frac{e_{36\,46\,15}}{X_{36}X_{46}}}$$

+0 $\begin{cases} 24 = 14 - 13 \\ 25 = -13 \\ 26 = -13 \end{cases}$ $BF = \begin{pmatrix} 35 & 36 \\ & 46 \end{pmatrix} \rightarrow e_{36\,46\,15} = 0$

+1 $\begin{cases} 35 = 25 - 24 \\ 36 = 26 - 24 \\ 31 = -24 \end{cases}$ $BF = \begin{pmatrix} 46 & 41 \end{pmatrix} \xrightarrow{\text{step 1}} e_{26\,46\,15} = 0$

+2 $\begin{cases} 46 = 36 - 35 \\ 41 = 31 - 35 \\ 42 = -35 \end{cases}$ $BF \begin{pmatrix} 52 \\ 62 \end{pmatrix} \rightarrow e_{26\,36\,15} = 0$

+3 $\begin{cases} 51 = 41 - 46 \\ 52 = 42 - 46 \\ 53 = -46 \end{cases}$ $BF \begin{pmatrix} 62 & 63 \\ & 14 \end{pmatrix} \nearrow$

---

next step take $\mathrm{Res}_{X_{26}}$ and check which constraints descend to $n=5$

+0 : $\begin{cases} 24 = 14 - 13 \\ 25 = -13 \\ 26 = -13 \end{cases} \rightarrow \begin{matrix} 24 = 14 - 13 \\ 25 = -13 \\ 13 = 0 \end{matrix}$ NO

+1 : $\begin{cases} 35 = 25 - 24 \\ 36 = 26 - 24 \\ 31 = -24 \end{cases} \rightarrow \begin{cases} 35 = 25 - 24 \\ 31 = -24 \end{cases}$ ✓

+2 : $\begin{cases} 46 = 36 - 35 \\ 41 = 31 - 35 \\ 42 = -35 \end{cases} \rightarrow \begin{cases} 41 = 31 - 35 \\ 42 = -35 \end{cases}$ ✓

+3 $\begin{cases} 51 = 41 - 46 \\ 52 = 42 - 46 \\ 53 = -46 \end{cases} \rightarrow \begin{cases} X_{52} = X_{42} - X_{41} \\ X_{53} = -X_{41} \end{cases}$ ✓

↓ saw this earlier

$$\underset{X_{26}}{\mathrm{Res}} = \cancel{\frac{e_{13\,15\,26}}{X_{13}}} + \cancel{\frac{e_{14\,15\,26}}{X_{14}}} + \cancel{\frac{e_{24\,15\,26}}{X_{24}}} + \cancel{\frac{e_{25\,15\,26}}{X_{25}}} + \cancel{\frac{e_{35\,15\,26}}{X_{35}}} \rightarrow 35 \text{ CF on } +1$$

BF on +3   BF on +1   24 CF on +3   24 BF on +2

Res at 36

$+0$: $\begin{cases} 24 = 14 - 13 \\ 25 = -13 \\ 26 = -13 \end{cases} \rightarrow \begin{cases} 24 = 14 - 13 \\ 25 = -13 \end{cases}$ ✓

$+1$: $\begin{cases} 35 = 25 - 24 \\ 36 = 26 - 24 \\ 31 = -24 \end{cases} \rightarrow \begin{cases} 35 = 25 - 24 \\ 31 = -24 \end{cases}$ ✓

$+2$: $\begin{cases} 46 = 36 - 35 \\ 41 = 31 - 35 \\ 42 = -35 \end{cases} \rightarrow \begin{cases} 41 = 31 - 35 \\ 42 = -35 \end{cases}$ ✓

$+3$: $\begin{cases} 51 = 41 - 46 \\ 52 = 42 - 46 \\ 53 = -46 \end{cases} \rightarrow \begin{cases} X_{52} = X_{42} - X_{41} \\ X_{53} = -X_{41} \end{cases}$ ✓

same steps as above show that $e_{15\,36\,\bullet} = 0$

$$\underset{X_{36}}{\mathrm{Res}} = \cancel{\frac{e_{13\,15\,36}}{X_{13}}} + \cancel{\frac{e_{14\,15\,36}}{X_{14}}} + \cancel{\frac{e_{24\,15\,36}}{X_{24}}} + \cancel{\frac{e_{25\,15\,36}}{X_{25}}} + \cancel{\frac{e_{35\,15\,36}}{X_{35}}} \rightarrow 35 \text{ CF on } +1$$

BF on +3   BF on +1   24 CF on +3   24 BF on +2

Res 46:

$+0$: $\begin{cases} 24 = 14 - 13 \\ 25 = -13 \\ 26 = -13 \end{cases} \rightarrow \begin{cases} 24 = 14 - 13 \\ 25 = -13 \end{cases}$ ✓

$+1$: $\begin{cases} 35 = 25 - 24 \\ 36 = 26 - 24 \\ 31 = -24 \end{cases} \rightarrow \begin{cases} 35 = 25 - 24 \\ 31 = -24 \end{cases}$ ✓

$+2$: $\begin{cases} 46 = 36 - 35 \\ 41 = 31 - 35 \\ 42 = -35 \end{cases} \rightarrow \begin{cases} 41 = 31 - 35 \\ 42 = -35 \end{cases}$ ✓

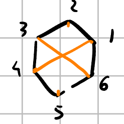
$+3$ ✗

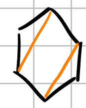
$$\underset{X_{46}}{\mathrm{Res}} = \cancel{\frac{e_{13\,15\,36}}{X_{13}}} + \cancel{\frac{e_{14\,15\,36}}{X_{14}}} + \cancel{\frac{e_{24\,15\,36}}{X_{24}}} + \cancel{\frac{e_{25\,15\,36}}{X_{25}}} + \cancel{\frac{e_{35\,15\,36}}{X_{35}}} \rightarrow 35 \text{ CF on } +1$$

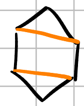
CF on +2   BF on +1   CF +0   24 BF on +2

→ any diagram with something crossing 15 vanishes. → by rotations, everything with something crossing a short vanishes.

next Res. at 14







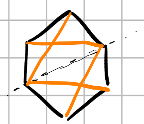
thing that cross (excluding starts): 25, 36

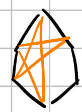
including starts: 62, 63, 52, 53

$$\mathrm{Res}_{14} = (----) + \frac{1}{25}\left(\frac{e_{14\,25\,15}}{X_{15}} + \frac{e_{14\,25\,24}}{X_{24}}\right) + \frac{1}{36}\left(\frac{e_{14\,36\,13}}{X_{13}} + \frac{e_{14\,36\,46}}{X_{46}}\right) + \frac{1}{25\,36}\, e_{14\,25\,36}$$

remove 62, 63, 53, 52, 14

→ leaves 64, 51, 24, 13





$$+0: \begin{cases} 24 = 14 - 13 \\ 25 = 15 - 13 \\ 26 = -13 \end{cases} \xrightarrow{14=0} \begin{cases} 24 = -13 \\ 25 = 15 - 13 \end{cases}$$

$$+1: \begin{cases} 35 = 25 - 24 \\ 36 = 26 - 24 \\ 31 = -24 \end{cases} \xrightarrow{14=0} 31 = -24$$

$$+2: \begin{cases} 46 = 36 - 35 \\ 41 = 31 - 35 \\ 42 = -35 \end{cases} \xrightarrow{14=0} \begin{cases} 46 = 36 - 13 \\ 42 = -13 \end{cases}$$

$$+3 \begin{cases} 51 = 41 - 46 \\ 52 = 42 - 46 \\ 53 = -46 \end{cases} \xrightarrow{14=0} \begin{cases} 51 = -46 \\ 52 = 42 - 46 \end{cases}$$

$$+4 \begin{cases} 62 = 52 - 51 \\ 63 = 53 - 51 \\ 64 = -51 \end{cases} \xrightarrow{14=0} 64 = -51$$

$$+5 \begin{cases} 13 = 63 - 62 \\ 14 = 64 - 62 \\ 15 = -62 \end{cases} \xrightarrow{14=0} \begin{cases} 13 = 63 - 64 \\ 15 = -64 \end{cases}$$

$\frac{1}{25\,36}\, e_{14\,25\,36}$ term:

+0 Gn here 36 BF and 25 CF

$$\mathrm{Res}_{X_{25}} = \frac{a_{14\,25\,15}}{X_{15}} + \frac{a_{14\,25\,24}}{X_{24}}$$

use (+1) $X_{31} = -X_{24}$ can choose 15 as independent $\Rightarrow$ $a_{14\,25\,15} = 0$

use (+4) $X_{64} = -X_{51}$ can choose 24 as independent $\Rightarrow$ $a_{14\,25\,24} = 0$

$$\mathrm{Res}\, X_{36} = \frac{a_{14\,36\,13}}{X_{13}} + \frac{a_{14\,36\,46}}{X_{46}}$$

(+1) $X_{31} = -X_{24}$ $\rightarrow$ X_{46} BF so $a_{14\,36\,46} = 0$

(+4) $X_{64} = -X_{51}$ $\rightarrow$ X_{13} BF so $a_{14\,36\,13} = 0$

$\rightarrow$ all non-locals vanish at $n=6$ with this method.

Next time: check $n=7$